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Comparative analysis of loan risk forecasting using quantum machine learning and classical machine learning models

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Research Article

Keywords: loan risk forecasting SVM QSVM Quantum

Posted Date: October 21st, 2025

DOI: https://doi.org/10.21203/rs.3.rs-7907390/v1

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Additional Declarations: The authors declare no competing interests.

Abstract

Non-performing loans present a significant challenge to financial institutions, driven by the complexity of the dataset, default probability, and default correlation(Bellotti et al., 2019). To mitigate this risk, this study investigates the potential of Machine Learning (ML) and Quantum Machine Learning (QML) algorithms for forecasting loan risk. Using a dataset from Kaggle, we conducted a comparative analysis between Support Vector Machine (SVM) and Quantum Support Vector Machine (QSVM).

Our result using a dataset of 12,368 records and 12 features shows that the QSVM model outperformed SVM, with a higher true positive rate (93.2.%) and true negative rate (87.6%), demonstrating better performance in identifying both default and non-default cases. Additionally, QSVM exhibits a lower false negative rate indicating its superior ability to minimize clients likely to default. The AUC score of 1.0 for the QSVM further demonstrates its exceptional ability in loan prediction. While the dataset used allowed for a solid comparison, QSVM demonstrated its capacity to continue improving with larger datasets, showing its scalability and strong potential application in loan risk forecasting especially with larger datasets.

1 Introduction

Lending is one of the primary responsibilities of banks and other financial institutions that is important to their profitability. However, unpredictability associated with loan repayment by borrowers is the primary source of risk for financial institutions to predict. Loan default can lead to significant bank losses, affecting financial institutions' profitability and stability(Arutjothi & Senthamarai, 2018). The issue of loan risk has been an important subject to banking and financial institutions, as it plays a crucial role in managing lending institutions' financial stability and profitability.

The origin of loan risk forecasting can be traced to the 1930s when selected mail-order businesses adopted such techniques to address inconsistencies among their loan assessment by personnel. Traditional loan risk models such as logistic regression, linear discriminant analysis, probabilistic data analysis, Monte Carlo prediction, etc. have been widely used in credit scoring applications due to their simplicity, and ease of implementation(Guskov & Levin, 2017). These models analyze borrower's historical data by gathering information such as income, financial statements, and other pertinent data.

Scoring techniques also known as scorecards are another method utilized to evaluate loan applications. The score aims to predict the likelihood of an applicant repaying the requested loan and assess the risk of potential defaults. A loan application scorecard is developed based on previous statistical data, categorizing an application as good or bad credit risk. Past loan applications are analyzed to identify the key factors significantly impacting the classification of creditworthy and high-risk borrows(Nehrebecka, 2018).

However, the relationship between the historical data of a borrower changes over time due to the change in economic conditions and market dynamics. Recent studies have shown the traditional model may not accurately capture the complexity of the modern financial market and borrower behavior, which can lead to deficiencies in loan risk assessment and an increased risk model failure necessitating more adaptive and

nuanced approaches to risk management that consider the unique circumstance of a borrower. Machine learning algorithms are being increasingly applied to loan risk forecasting. This model has been shown to perform better than the traditional loan risk model and can process high-dimensional data(Leo et al., 2019).

The emergence of Quantum computing has attracted significant attention in various fields, including finance. Quantum machine learning is at the nexus of machine learning and quantum computing. The primary objective of quantum machine learning is to speed up processes(Zeguendry et al., 2023). Quantum machine learning algorithms such as Quantum Support Vector (QSVM), Quantum Neural Networks (QNN), Variational Quantum Classifier, Quantum Boltzmann Machine (QBM), etc. have demonstrated the potential to outperform their classical counterpart.

To ensure good cooperate governance in banking there is a critical need for novel ways to offer quicker, more accurate, and scalable solutions as financial institutions struggle with the complexity of risk assessment and management. We leverage the potential of ML and QML to improve the predictiveness of loan risk forecasting. ML and QML offer a viable approach to unlock new insight and improve the accuracy of risk predictions. With the inherent parallelism and probabilistic nature of quantum systems, these algorithms can potentially uncover complex patterns and relationships within the financial data that elude classification approaches.

In this study, we implement a robust predictive model using ML and QML techniques, aiming to identify the most effective approach for predicting potential defaulters. Additionally, the research explores how QML can improve loan risk forecasting by leveraging the parallelism and probabilistic properties of quantum systems. The study offers a comparative analysis between ML and QML methods in loan risk prediction, assessing their respective performance in terms of accuracy.

1.1 Research Objectives

This study hopes to give financial institutions a more dependable and effective tool to enhance their decision-making process, optimize their lending portfolio, increase capital adequacy ratio, and stay ahead of the curve in the ever-changing financial landscape future evaluating loan risk, empowering them to decide on loans more wisely and reduce losses, this will lead to economic growth and stability of a country. To achieve this goal our objectives are as follows:

- To implement SVM and QSVM to predict the default probabilities of loan
- To Perform a comparative study evaluating the performance of SVM and QSVM models in predicting loan risk.

1.2 Paper Structure

The rest of this paper is structured as follows: Section 2 reviews the key concept of machine learning and quantum computing, Section 3 section details the experimental methodology, tools, and models, section 4

presents results, highlighting QSVM's advantages, section 5 discusses findings, challenges, and contributions, emphasizing quantum machine learning's potential in financial forecasting

2 Literature Review

The issues of loan risk have been the subject of extensive study in banking and financial institution, as it plays a major role in managing lending institution's financial stability and profitability. Accurately forecasting loan risk is important for financial management, especially for banking and lending institutions, this process involves predicting the probability of a borrower defaulting on a loan, enabling financial institutions to mitigate potential losses and effectively manage it.

Several studies have been conducted using ML to solve loan risk prediction, such studies include(Alsaleem & Hasoon, 2020) that analyze the performance of various ML algorithms, such as multilayer perceptron, random forest, BayesNet, NaiveBayes, and DTJ48, in classifying bank loan risk. The author utilized a dataset comprising 1,000 instances and 11 attributes and found that the multilayer perceptron algorithm outperformed the other algorithm. However, the relatively small size of the dataset raises concerns about the model's ability to generalize effectively, as it may have simply memorized the training data rather than learning the underlying patterns. A relatively small dataset may not adequately capture the complexity of the problem, potentially leading to biased predictions. Additionally, the author did not consider other critical factors, such as the financial implications of false-positive and false-negative decisions in the context of loan approval, which have significant consequences for banks and financial institutions.

The author (Venkata Lakshmi Dharani & Thanuja, 2021)utilizes various machine learning algorithms, including the KNN classifier, Random Forest Classifier, Decision Tree, and Logistic Regression, to analyze the lending club dataset. The paper highlights the challenges of the imbalance dataset used, which is a common issue in loan datasets where the number of defaults (bad loans) is significantly lower than the number of repayments (Good loans). The large disparity between good loans (92.21%) and bad loans (7.7%) can result in the model learning noise, making accurate prediction more difficult. Although the author employs techniques like SMOTE (Synthetic Minority Oversampling Technique) and NearMIss to improve the reliability of their prediction. However, while these methods aim to balance the dataset, their effectiveness may vary, particularly in predicting the minority class (bad loans. SMOTE generates synthetic examples by interpolating between existing minority class samples, but if those samples contain noise or outliers, it can lead to overfitting, As a result, the model may perform well on the training data but struggle to generalize to unseen data, as it may have memorized the synthetic example rather than learning broader pattern. The author could have considered alternative approaches such as ADASYN, Tomek links, or cluster-based over-sampling, which mitigates these challenges depending on the specific characteristics of the dataset and the problem at hand.

Purity Monje Mwalozi (Mwalozi, 2023)develop a machine learning model to predict the likelihood of loan default. It utilized approximately 85,00 loan records from loans disbursed between January 2018 and January 2023, focusing on SACCOs in Kenya, particularly Okoa Management Ltd. The research employs various machine leaning algorithms, including logistic Regression, Decision, and Trees to train the model

on SACCOs data. Logistic Regression emerged as the most accurate model, achieving a prediction accuracy of 83% for loan default. Decision Tree also performs well, with an accuracy of 90%, especially in identifying key variables like home ownership and credit issues as strong prediction defaults. Given that many borrower characteristics and loan default tendencies exhibit non-linear relationships, the study could have further leveraged non-linear models such as Decision Trees and Support Vector Machines to capture more complex interactions among variables.

Gu utilized 5,000 loan datasets to provide a thorough analysis of their model performance, including detailed parameter setting and a comparison with other models like random forest and support vector machines. The result indicated the LSSV model has higher accuracy and real rate. The model achieved an accuracy of 90.15% and a recall of 85.63%. However, the study acknowledges challenges related to an unbalanced data sample and a limited amount of data which could affect the robustness and generalization of the model's prediction(Gu, 2022).

Larger datasets play a key role in improving the performance and reliability of machine learning models. Small datasets may cause the model to perform well on the training data but poorly on unseen data. A Study(Mestiri, 2024) was conducted using 688 datasets to develop a model for categorizing credit applicant to predict their likelihood of default. The author evaluated several classification algorithms which include Linear Discriminate Analysis (LDA), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF) and Deep Neural Network (DNN). Although the DNN provided the highest score, the study's results could have been improved if a larger dataset had been utilized, allowing for better generalization and more robust model performance

The study (Vravikumar et al., 2022.)utilized a dataset of 1500 records, divided into 80% for training and 20% for testing. Logistic regression was employed as the primary method for predicting loan approval and establishing a statistical modeling technique. Additionally, Random Forest was implemented for comparison. The dataset comprised data from previous clients of multiple banks, with loan approval based on criteria such as credit history, income, and loan amount. The model achieved an accuracy of 81.1% in predicting loan approval status. However, relying heavily on logistic regression may limit the ability to capture the complex, nonlinear relationships between input features and loan approval outcomes more advanced model such as SVM or deep learning, could have improved the study. Moreover, the relatively small dataset (1500 samples) may limit the model's ability to generalize which may lead to overfitting.

The paper(Danenas & Garsva, 2012) introduces a hybrid approach that combines Support Vector Machine classifiers with evolutionary optimization. This combination enables the automatic selection of optimal parameters, thereby improving model performance compared to manually selected parameters. Furthermore, the author applied a sliding window approach to continuously update the model based on new data, making it more dynamic and adaptive to changing conditions in credit evaluation. The GA-LinSVM and PSO-LinSVM classifiers exhibit superior accuracy compared to the standard model, achieving over 88% accuracy. However, the paper acknowledges the challenge of imbalanced data and suggests that they could benefit from a more detailed exploration of techniques such as oversampling or cost-sensitive

learning to address the issue. Table 1 presents a summary of the review, methodology used, gaps, and opportunities for future research.

In summary, Some of the above studies rely on relatively small datasets, limiting the generalization of their result. To improve the robustness of findings, future research should focus on larger and more complex financial datasets. While techniques like SMOTE are commonly used to address imbalanced datasets, they can lead to issues such as overfitting. More advanced methods including quantum models like QSVM, QNN, Qk-NN, and VQC could provide better solutions for handling imbalanced data. Additionally, much of the existing research is centered on classical models like logistic regression, decision trees, and SVM, which often struggle with the non-linear complexities of financial data. There is a significant gap in exploring the potential of quantum and non-linear models such as Quantum support vector Machine (QSVM), Quantum Neural Network (QNN), Quantum K-Nearest Neighbors (Qk-NN), and Variation Quantum Classifier (VQC) to achieve better accuracy.

Table 1
Summary of literature review

Author/Date	Methodology	Findings	Limitations/Gaps	Opportunity for Future Research
Alsaleem &Hasoon (2020)	Comparison of Various ML algorism (Multilayer perceptron, Random Forest, BayesNet, NaiveBAyes, DTJ48) on loan risk classification	Multilayer perceptron outperformed another algorithm	A small dataset (1000 instances) raises concerns about generalization and overfitting; no consideration of the financial implication of false positive/negative	Use larger, more complex datasets and investigate the financial consequences of misclassifications in loan approval decisions
Venkata Lakshmi Dharani & Thanuja (2021)	Comparison of ML algorithm (KNN, Randon Forest, Decision Tree, Logistic Regression) using an imbalance Lending Club dataset	Techniques like SMOTE were applied to improve the prediction of minority class (bad loan)	SMOTE can lead to overfitting due to synthetic samples; limited exploration of alternative techniques like ADASYN, Tomek link	Explore alternative resampling techniques, evaluate models' ability to generalize to unseen data, and compare another quantum model
Reddy et al (2020)	Analysis of lending club dataset using Decision Tree, Random Forest, AdaBoost, Gradient Boost, and Bagging Classifier	The Bagging Classifier achieved the highest performance (89% accuracy, Weighted F1 score of 0.75)	Lack of exploration of non-linear models and advanced techniques like quantum model	Investigate the effectiveness of QSVM in handling complex non-linear relationships in loan risk datasets.
Mwalozi (2023)	Prediction of loan default using logistic regression and decision tree on SACCOs data in Kenya	Logistic regression and decision trees achieved an accuracy of 83% and 90% respectively	Didn't explore more advanced method line SVM or non-linear model; limited examination of non-linear borrower characteristics	Explore non-linear model like SVM or QSVM for improved handling of borrower characteristics and default probabilities
Gu (2022)	Comparison analysis of LSSVM, random forest and support vector machine on loan risk data	LSSVM achieved 90.15% accuracy and 85.15 recall	Limited dataset (5,000 instances) unbalance data which may impact model robustness	Address unbalanced data with advanced techniques (eg. Oversampling, cost-sensitive learning) and explore quantum algorithms for better generalization

Author/Date	Methodology	Findings	Limitations/Gaps	Opportunity for Future Research
Danenas & Garsva (2012)	Hybrid SVM and evolutionary optimization for credit assessment	A hybrid approach with GA-LinSVM and PSO-LinSVM achieved 89% accuracy	Imbalanced data issues; limited exploration of advanced resampling techniques like cost-sensitive learning	Investigate QSVM's potential in a hybrid model and cost-sensitive approach for an imbalance credit dataset
Vravikumar, (2022)	Logistic Regression and Random forest on 1500 dataset for predicting loan	Maximum accuracy of 81.8% with logistic regression	Small dataset (1500 samples), the limited ability of logistic regression to capture the non-linear relationship	Explore non-linear models like SVM or deep learning for improved insight, use larger dataset for better generalization
Mestiri (2024)	Evaluated LDA, LR, DT, RF, DNN for loan default prediction	DNN provided the highest accuracy	Relatively small data (688)	A larger dataset needs to improve robustness and performance

3 Methodology

The method for this work involves selecting and applying some techniques to ensure accuracy and reliability in predicting loan defaults. This section outlines the process and methods used to analyze and transform borrower data, implement predictive models, and evaluate performance.

3.1 Dataset Description

This study was conducted using a dataset downloaded from Kaggle.com. the dataset consists of 12,638 rows and 12 columns, with both numeric and categorical attributes, and aims at conducting a comparative study in predicting loan risk. The dataset includes demographic and financial variables related to loan applications. These include the age of the individual, ranging from 20 to 144 years, and their income. The length of employment ranges from 0 to 123 years. The loan amount requested falls between $500 and $35,500, with interest rates ranging from 5.42% to 23%. The dataset also includes a binary indicator of whether the loan could be (default 1) or default and (0 default) for default as well as the percentage of the applicant's income that the loan amount represents. Which ranges from 0 to 78%. Finally, the loan history length in a year varies from 2 to 30 years. Figure 1 shows the summary of the dataset, with variables related to loan risk forecasting.

3.2 Data Processing

Data processing is critical in implementing an accurate and reliable loan risk forecasting model. The raw data shown in Fig. 1 include the borrower's attribute attributes such as age, income, home, intent, grade, employee length of work, loan interest rate, loan amount, and loan status. However, these data are

incomplete, imbalanced, and contain both numerical and categorical data. The processing phase plays an important role in transforming this raw, unstructured data into a structured, clean, and balanced dataset suitable for predicting loan default probabilities.

3.2.1 Data cleaning and Feature encoding

The dataset was uploaded in the Jupiter environment and examined to find the inconsistency related to the dataset set. Some missing values were identified in the dataset. we address the missing values by apply the mean imputation strategy, meaning any missing values are replaced by the mean of the respective column. For categorical we use the most frequent (mode) imputation strategy. That is replacing the missing entries with the mode. After handling the missing values, the categorical variable variables are also encoded into numerical using label encoding techniques.

3.2.2 Feature Scaling

The feature in the study contains income, loan amount, interest rate, and loan percentage income that vary widely in scale. To ensure all features like loan amounts do not dominate loan scores the study employs the Normalization technique also known as Min-Max Scaling. This ensures all features contribute equally to the model to avoid overfitting and numerical instability

3.3 Hyperparameter Tuning

For the SVM model, the chosen kernel was the Radial Basic Function (RBF) kernel, which maps the input space into a higher-dimensional space, making the data linearly separable. In the quantum version of the experiment, the hyperparameter includes the number of qubits, the generalization parameter, the random seed, the quantum backend, the feature map, and the quantum kernel.The number of qubits corresponds to the dimensionality of the feature set. Each qubit represents one feature. Since the dataset has 12 features, 12 qubits were used in the quantum circuit. While increasing the number of qubits can capture more complex patterns, it also leads to higher computational demand. The regularization parameter, C, was used to balance the trade-off between maximizing the decision margin and minimizing classification error. In this case, C was set to 10 to reduce misclassification during training

Quantum_instance was configured to use the statevector_simulator backed from Qiskit, which simulates quantum circuits on a classical computer. Using the statevector_simulator ensures a noise-free environment. The ZFeatureMap was used to encode classical input data into a quantum state via parameterized quantum gates. Two repetitions of the feature mawp were applied to encode the feature through multiple layers of quantum operations, which increased the model's capacity to capture complex patterns. The quantum kernel is a matrix that measures the similarity between input data points using quantum states. This is based on the ZFeatureMap and runs on the configured quantum backend. The Quantum Support Classifier (QSVC) is the quantum equivalent of the classical SVM. The regularization

parameter was also applied to QSVC to manage the margin between classes and improve classification performance

3.4 SVM and QSVM Approach

The implementation of a support vector machine model for this study began with identifying and examining the dataset. The dataset was loaded onto the jupyter notebook environment. The data is then cleaned to ensure missing data are removed which may affect the performance or the work. The data was encoded to convert categorical features into numerical values using a label encoding technique since SVM cannot handle categorical data. The dataset was then split into 80% training and 20% testing, this will enable a thorough evaluation of the model's generalization capabilities. The Radial Basic function (RBF) was used to handle non-linear separable data. Data was then normalized using the MinMaxSCaler technique, which contains a value between 0 and 1, thereby mitigating the impact of varying feature magnitudes. The model was then trained, and a classification report, confusion matric report, and other metrics were generated.

Similarly, the implementation of QSVM in forecasting process began with the preparation of data for model evaluation. The dataset was divided into training and testing subsets using 80% training and 20% testing. This ensures that the model is trained on one portion of the data and evaluated on another to measure its generalization performance. Data was then normalized using the MinMaxSCaler technique, which contains a value between 0 and 1, thereby mitigating the impact of varying feature magnitudes. Twelve (12) qubits were you and the regularization parameter was set to 10. The QSVM is then trained using quantum kernel and ZfeatureMap which transform classical data into a quantum state. Data was then evaluated, and the relevant metrics were used for analysis.

3.5 SVM and QSVM Model Evaluation

1. The following metrics were used to evaluate the performance of the SVM and QSVM model,
2. True Positive (TP): This is the correctly predicted positive case (default loan)
3. True Negative (TN): This is the correctly predicted negative case (non-default loan)
4. False Positive (FP): This is the misclassification of default loans when they are non-default loan
5. False Negative (FN): This is the misclassification of non-default when they are default loan
6. Accuracy: measures the proportion of correctly predicted instances of default loan and non-default loan shown in Table 2 and Table 3
7. Precision: measure the proportion of correctly predicted non-default loans out of the loans predicted as positive
8. Recall: measure the proportion of correctly predicted non-default loans out of all actual non-default loan instances
9. F1- Score: This is the harmonic mean of precision and recall, balancing the two.

10. Area Under the Curve (AUC) is the performance measurement for the classification model, specifically related to the Receiver Operating characteristics (ROC) curve, the rocking curve plots the true positive rate against the false positive rate at various threshold levels.

3.6 Experimental Workflow

Figure 2. Shows the summary and the steps involved in the training and prediction process using SVM and QSVM. The following steps were taken to implement the study, ensuring a systematic approach to data preprocessing, feature selection, model training, and evaluation. This comprehensive workflow highlights the method used to assess the predictive capacities of classical and quantum machine learning models in loan risk forecasting

4 Results

This section presents the training and inference result of loan risk forecasting using a Support Vector Classifier and Quantum Support Vector Machine. There are several performance metrics used to evaluate the performance of SVM and QSVM. This study focuses on a few performance metrics.

4.1.1 Learning Curve SVM
The learning curve in Fig. 3 shows the training and cross-validation accuracy of the SVM model used for predicting the default probability of the loans, in relation to the number of training samples. The training accuracy started high and continues to rise as more samples are added, indicating that the model is effectively learning from the training data.

The training score stabilizes around 0.89, suggesting the model has reached its full capacity for fitting the training data. Meanwhile, cross-validation accuracy begins lower, which is expected, as the models generally perform better on the data they were trained on to unseen data. However, as the number of training samples grows, the cross-validation accuracy improves steadily. The cross-validation score level off around 0.855, suggesting the model reaches a point where further data doesn't substantially improve its generalization ability. The model benefits from more data, as evidenced by the steep rise in cross-validation accuracy. Eventually, the performance level off, and both curves flatten, indicating the model has learned as much as it can from the data and additional data does not lead to further improvement. The curve plateaus around 8,000 samples, also suggesting this number of samples is enough for the SVM model to achieve near-optimal performance on this task.

4.1.2 Confusion Matrix SVM
Figure 4 shows the confusion matrix where 0 indicates the probability of default and 1 indicates the probability of non-default. The model accurately identifies 1,176 of the loan records as defaults and correctly classified them. However, it misclassified 96 as non-default when they were defaulted.

On a positive note, it correctly predicted 1,095 loan records as non-default. This highlights the model's strong ability to predict non-default loans, with a precision of about 91.9%. However, the recall is slightly

lower with approximately 87.2% of actual non-defaults correctly identifying.

The model shows a strong recall for defaults, and correctly identifies 92.5% of actual defaults, indicating its effectiveness in detecting potential defaulters. While the model's overall accuracy is 90% shown in Table 2, there is a small trade-off between false positives and false negatives. This suggests that the model makes fewer mistakes when identifying defaults but may overlook some actual non-defaults.

4.1.3 Receiver Operating Characteristics (ROC) Curve for SVM

The SVM model strongly predicted loan risk, with an Area Under the Curve (AUC) of 0.898. The steep curve in Fig. 5 rise indicates that the model has a high true positive rate, accurately identifying a substantial proportion of clients who are likely to default. Additionally, the model demonstrates a moderate false positive rate, meaning it incorrectly classifies non-default clients as defaulting at a manageable level, showing reasonably robust performance. The AUC of 0.898 suggests that the model is quite effective, though not perfect, in distinguishing between clients who will default and those who will not.

Table 2
classification report for SVM

	Precision	**Recall**	**F1 Score**	**Support**
0	0.88	0.92	0.90	1272
1	0.92	0.87	0.89	1256
Accuracy			0.90	2528
Macro avg	0.90	0.90	0.90	2528
Weighted avg	**0.90**	**0.90**	**0.90**	**2528**

4.2 Performance of Quantum Support Vector Machine

4.2.1 Learning Curve QSVM

Figure 6 presents the learning curve for QSVM in predicting loan risk. Initially, the training accuracy is quite high, around 0.93, when using smaller datasets. This shows that the model is fitting the training data well at this stage. However, as more training samples are introduced, the accuracy gradually decreases leveling off around 0.91. this slight dim suggests that with more data, the model encounters more complex patterns, causing a small drop in its ability to fit the training data. This drop shows the QSVM tendency to overfit smaller datasets as it performs exceptionally well at first, but the benefit diminishes as more data is added.

On the other hand, the cross-validation accuracy starts relatively low, around 0.87, showing the model's initial difficulty generalizing with limited data. As the training sample size grows, the cross-validation accuracy steadily improves, reaching around 0.90 with 8,000 samples. This indicates that the model benefits scientifically from more data, enhancing its ability to generalize and predict new unseen data more accurately. AS the cross-validation accuracy approaches the training accuracy as the dataset grows shows that the model is learning to generalize better, reducing its initial overfitting

Initially, the gap between the training and cross-validation score is wide, suggesting the QSVC is overfitting the training data and struggling to generalize to unseen data. As the dataset increases, the gap closes, which is a good show that the model is balancing out and reducing overfitting by encountering more diverse data.

The shaded area around the trading and cross-validation curves represents confidence intervals. When the data set is small, this region is wider, indicating higher variability in performance. As more data is introduced, especially for cross-validation, the variability decreases, showing the model's performance stabilizes as the data size grows.

The QSVC shows strong performance, with both training and validation set data size increase, with the cross-validation scores catching up to training scores. The steady improvement in cross-validation suggests that the QSVC is well suited for the loan default probability prediction, particularly with more datasets.

Table 3
classification report for QSVM

	Precision	**Recall**	**F1 Score**	**Support**
0	0.88	0.93	0.91	1272
1	0.93	0.88	0.90	1256
Accuracy			0.90	2528
Macro avg	0.91	0.90	0.90	2528
Weighted avg	**0.91**	**0.90**	**0.90**	**2528**

4.2.2 Confusion Matrix for QSVM

Figure 7 presents the confusion matrix of the QSVC model used for the loan risk prediction. In this class 0 represents the probability of loan default, while 1 indicates the probability of non-default.

The model demonstrated a high accuracy in identifying default loans, with 93.2% of the loans predicted as default being accurately classified as such. Out of the 1272 actual default loans, the QSVC model correctly predicted 1186, resulting in a false positive rate of 6.8%. This means that 6.8% of loans were incorrectly classified as safe when they risky. Again, the model had a lower rate of misclassifying safe loans as risky, with only 12.4% of safe loans being wrongly predicted as risky. However, the model missed 156 safe loans.

The false positive percentage, which indicates the model's false alarm rate, is relatively low, showing a high level of specificity. Furthermore, the model demonstrates a strong true negative rate, correctly identifying 87.6% of the loans predicted as non-default.

In conclusion, the QSVM performs well in classifying loan default and no-default probabilities, with strong accuracy in detecting default.

4.2.3 Receiver Operating Characteristics (ROC) Curve for QSVM

Figure 8 presents the ROC curve of the QSVM model for loan risk forecasting. It shows a perfect performance. The model accurately identifies positive instances, with the curve reaching the top-left corner, indicating a true positive rate of 1.0 across the board. Conversely, the model does not incorrectly predict positive outcomes when the actual outcome is negative, as the false positive rate remains at 0, suggesting that the model perfectly distinguishes between default and non-defaults.

An AUC (area under the curve) score of 1.0 indicates that the QSVM model can perfectly separate the two classes (default 0, and non-default 1) without minimal error. This implies that the QSVM model is highly effective in predicting loan risk, as it can distinguish between clients who will default and those who will not default.

4.2.4 Comparative Analysis

This study aims to compare the performance of SVM and QSVM models in loan risk forecasting. The results indicate that the QSVM exhibits marginally improved performance compared to the SVM model in predicting loan default. QSVM maintains a true positive rate of 93.2%, slightly exceeding SVM's 92.5%, indicating QSVM is slightly better at accurately identifying clients who are likely to default. Furthermore, QSVM demonstrates a true negative rate of 87.6%, which is also slightly superior to the SVM. While both models perform well, QSVM outperforms SVM in correctly predicting default. QSVM additionally showcases a lower false positive rate than SVM, suggesting QSVM is slightly more effective in minimizing the misclassification of default clients as non-defaulting. Similarly, QSVM exhibits a slightly lower false negative rate compared to SVM, indicating QSVM is marginally better at accuracy in identifying clients who are likely not to default. The AUC of 0.898 suggests that the SVM model generalizes well and could perform effectively across different datasets, though it does not achieve the perfect classification of the QSVM of 1.0. The QSVM provides an excellent balance between identifying default risk and minimizing misclassification errors.

5 Discussion and Practical Significance of the Study

This study's findings lie in the advantage QSVM has over traditional SVM in predicting loan risk. The results from Table 3 indicate that QSVM can significantly enhance risk assessment in real-world financial settings. with a higher true positive rate (93.2.%) and true negative rate (87.6%), QSVM proves to be more effective in identifying high-risk loans. This improvement allows financial institutions to make more informed decisions during loan approval, reducing the chance of loan default and improving the dependability between high-risk and low-risk loans, minimizing false positives and false negatives is important for reducing financial losses.

While the SVM model often struggles with nonlinearities present in financial data, such as borrower behavior and market volatility, QSVM offers a more advanced approach to capturing the complexity,

resulting in more accurate decision boundaries for loan classification. This capability is valuable in today's financial landscape, where datasets are becoming more intricate, and more advanced models are required to handle them effectively.

Although quantum computing is still in the early stages, this study shows that QSVM can efficiently manage high-dimensional financial datasets. This efficiency is important for large financial institutions that deal with vast data related to loans, aging analysis, and collateral risk analysis. QSVM's ability to scale and process larger datasets could transform financial management, making risk model more adaptable and responsive

In conclusion, QSVM's performance in loan risk prediction has wide-reaching implications. It suggests a new era in risk management for the financial sectors, where quantum-enhanced models offer a more accurate, efficient, and reliable assessment of financial risk. QSVM's ability to handle complex data patterns and produce classification out has the potential to change loan risk assessment, loan, optimize loan approval process, and reduce financial risk for lenders.

5.1 Summary of the Findings

This study compared the performance of SVM and QSVM models in predicting loan risk default probability. The results show that QSVM achieved superior accuracy and performance in loan prediction compared to the SVM model. The QSVM model demonstrates slightly superior performance, with a true positive rate of 93.2%, slightly better than SVM's 92.5%, indicating it improves accuracy in identifying clients likely to default. QSVM also has a higher true positive of 87.6% compared to SVM, meaning it more accurately identifies non default. Additionally, QSVM exhibits lower false positive and false negative rates, suggesting it is more effective at minimizing misclassification in both directions.

Furthermore, QSVM shows the capacity to improve its performance as the dataset size increases, suggesting it could potentially perform even better with larger datasets. An AUC of 1 compared to the SVM of 0.898, provides an excellent balance between identifying default risk and minimizing misclassification.

In conclusion, while both models perform well, QSVM offers marginally improved accuracy over SVM in predicting non-default and default clients. This aligns with the existing research on loan risk forecasting, highlighting the potential of quantum-enhanced models like QSVM to improve financial risk prediction, especially with more extensive datasets.

5.2 Contributions

One of the contributions of this study is the successful application of QSVM in the context of loan risk forecasting. This study is among the few that have compared the effectiveness of SVM to QSVM and demonstrated the viability of QSVM in solving real-world financial problems, showing how quantum algorithms can outperform classical algorithms.

Declarations

This research received no funding or grant from any funding agency in the public, commercial, or non-profit organizations. The Authors also declare no competing interests.

Conflict of Interest

The authors have no relevant financial or non-financial interest to disclose.

Data Source

https://www.kaggle.com/datasets/laotse/credit-risk-dataset

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Figures

	age	income	home	emp_length	intent	grade	loan_amnt	loan_int_rate	loan_status	loan_percent_income	default	cb_person_cred_hist_length
0	22	59000	3	123.0	4	3	35000	16.02	1	0.59	1	3
1	25	9600	0	1.0	3	2	5500	12.87	1	0.57	0	3
2	23	65500	3	4.0	3	2	35000	15.23	1	0.53	0	2
3	24	54400	3	8.0	3	2	35000	14.27	1	0.55	1	4
4	21	9900	2	2.0	5	0	2500	7.14	1	0.25	0	2
...	...	...	...	...	...	...	...	...	...	...	...	...
14414	57	53000	0	1.0	4	2	5800	13.16	0	0.11	0	30
14415	54	120000	0	4.0	4	0	17625	7.49	0	0.15	0	19
14416	65	76000	3	3.0	2	1	35000	10.99	1	0.46	0	28
14417	56	150000	0	5.0	4	1	15000	11.48	0	0.10	0	26
14418	66	42000	3	2.0	3	1	6475	9.99	0	0.15	0	30

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Figure 1

Dataset for SVM and QSVM experiment

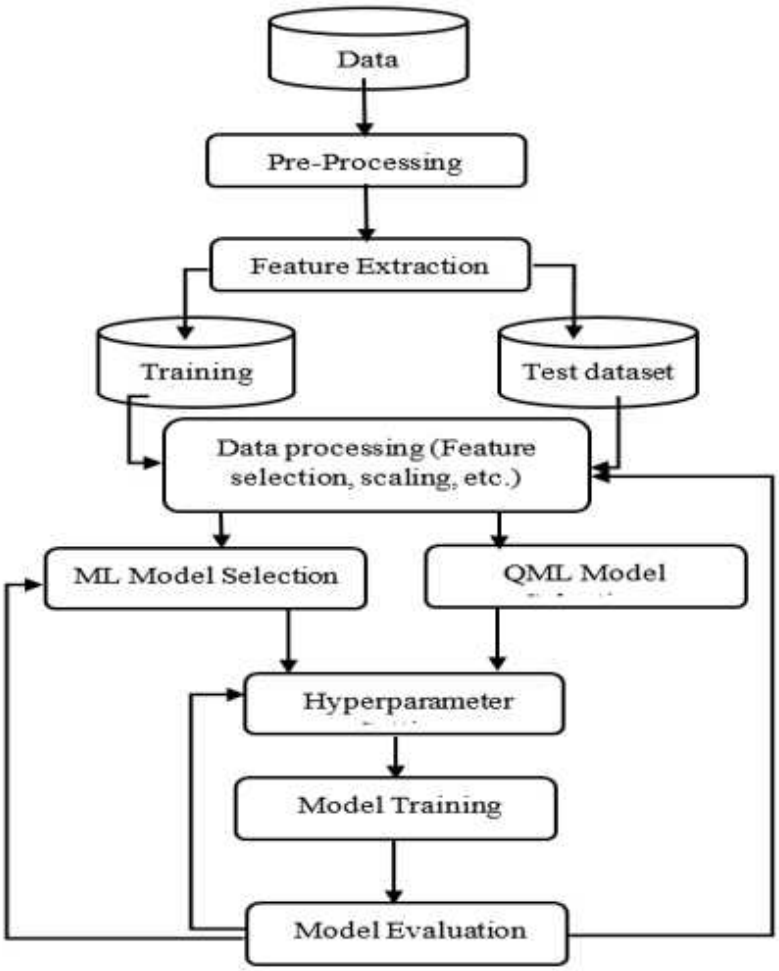

Figure 2

Experimental workflow of SVM and QSVM

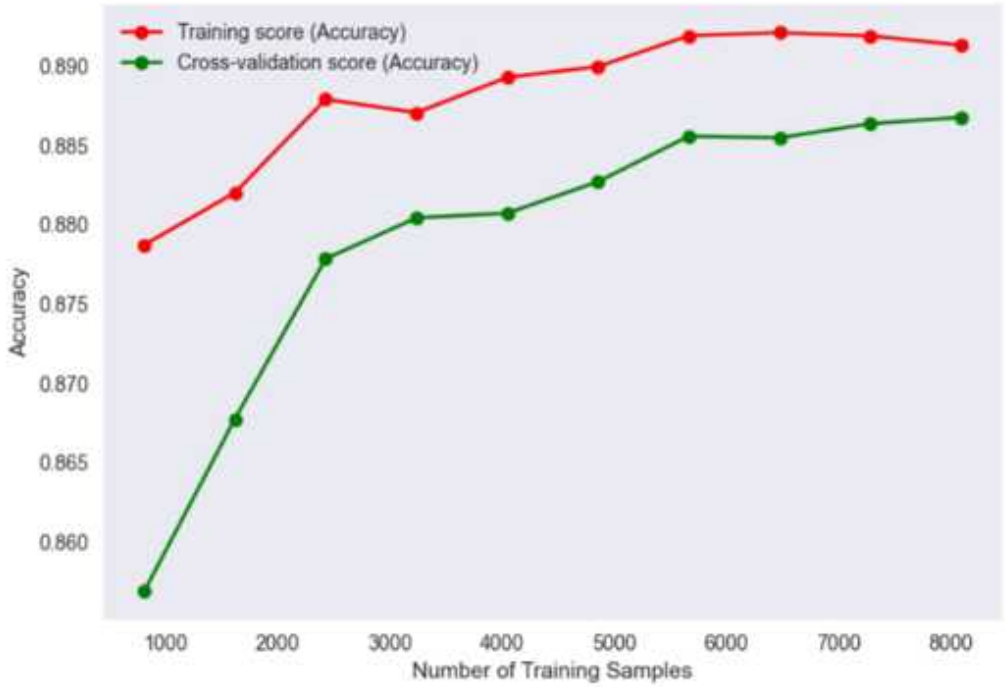

Figure 3

Learning Curve of Support Vector Machine

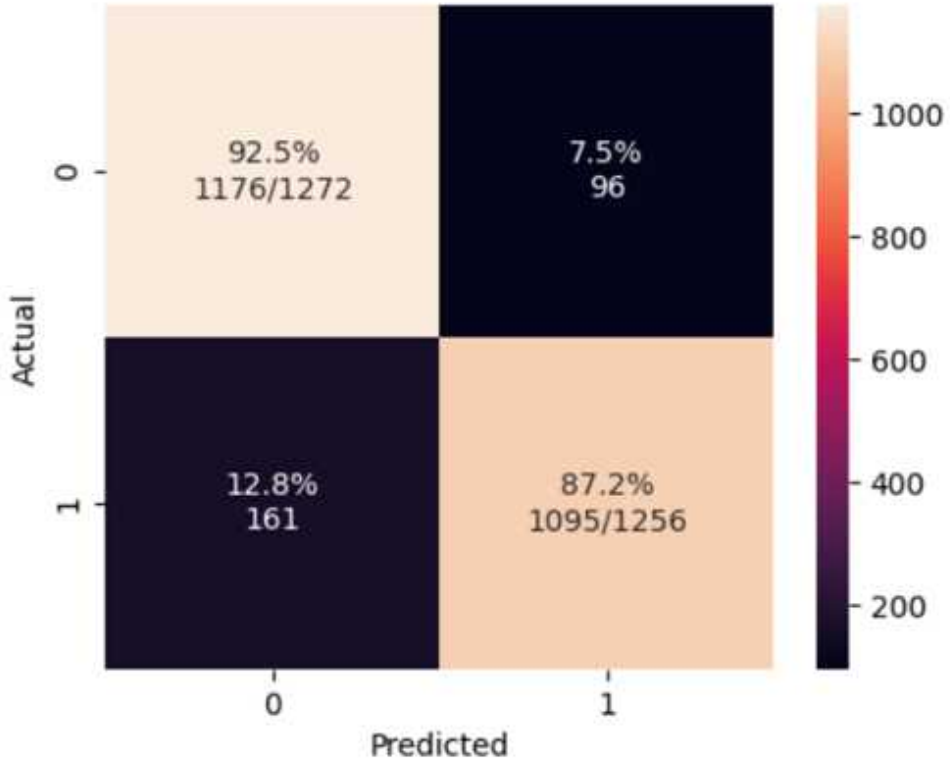

Figure 4

Confusion Matrix of Support Vector Machine

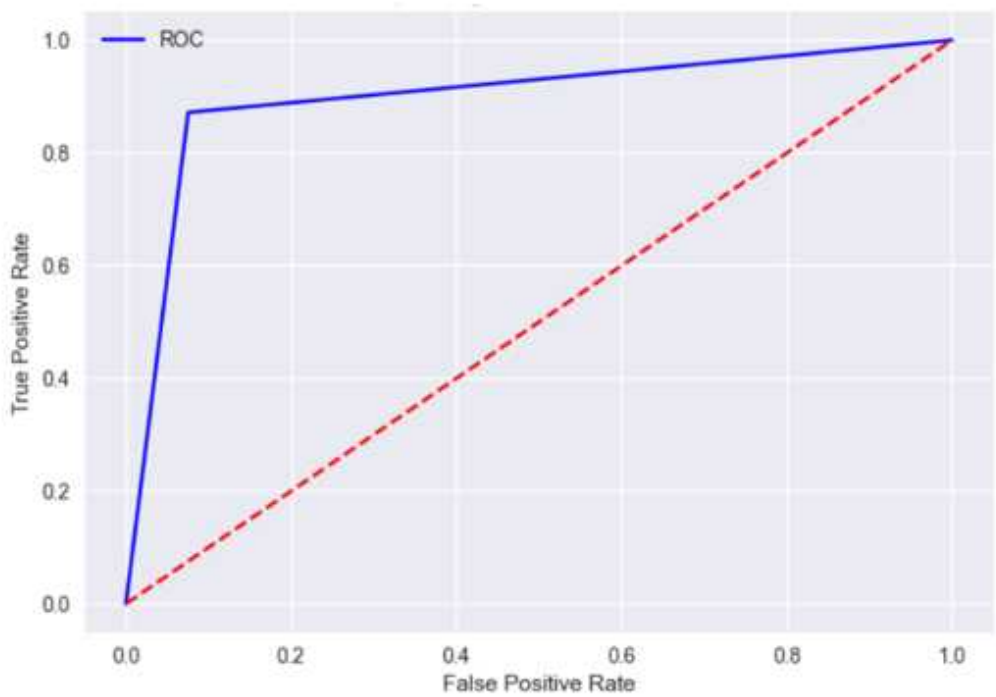

Figure 5

Receiver Operating Characteristics (ROC) Curve for Support Vector Machine

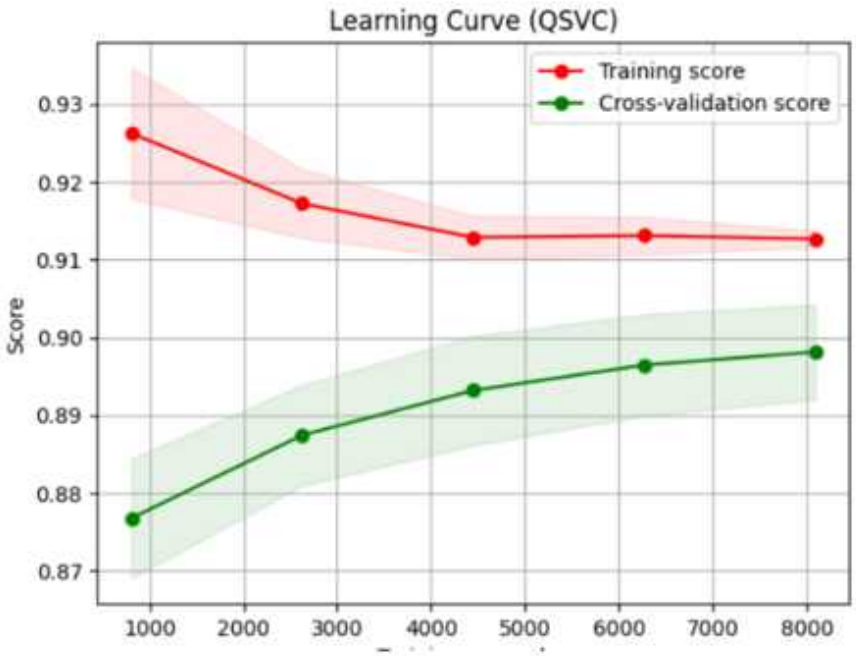

Figure 6

Learning curve for Support Vector Machine

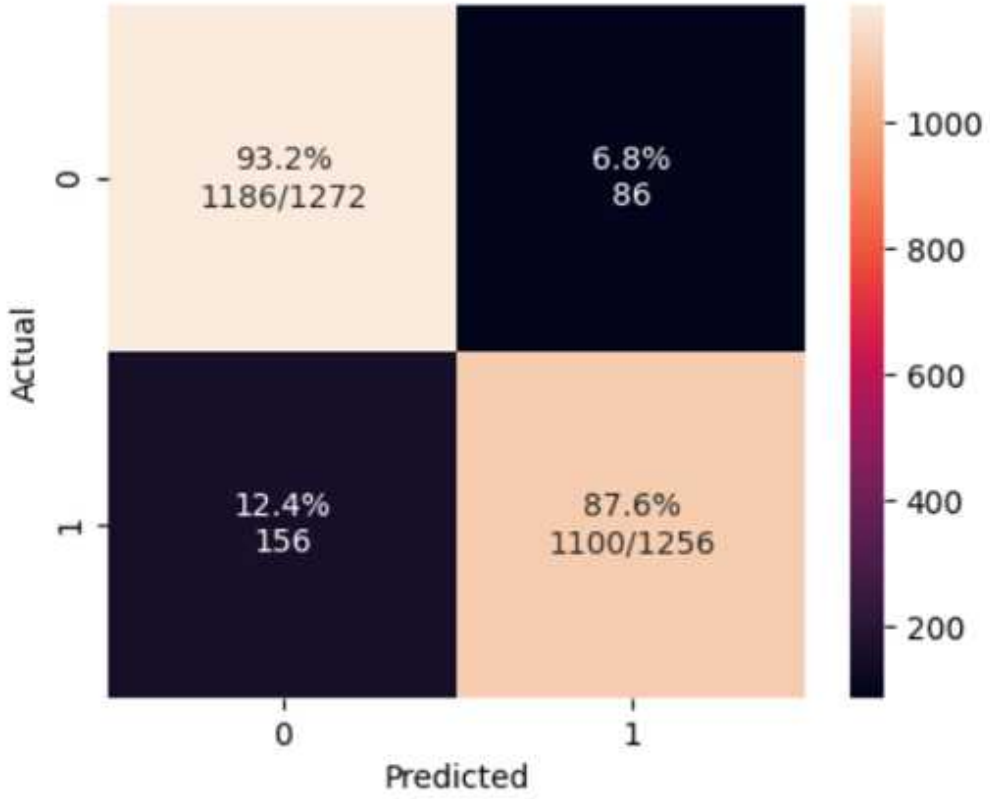

Figure 7

Confusion Matrix for Support Vector Machine

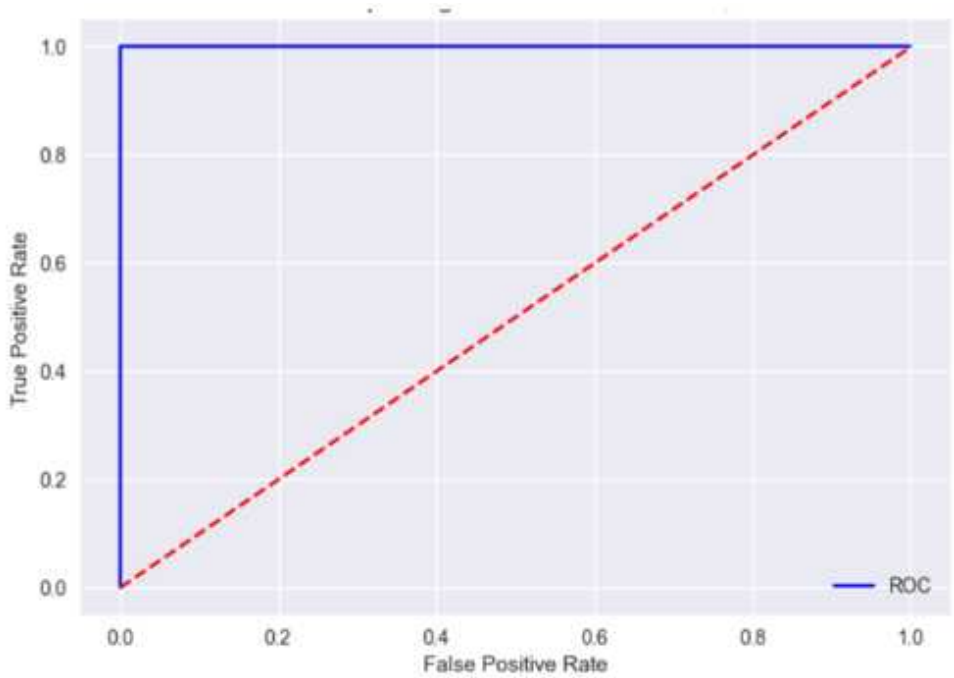

Figure 8

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- AppendixAVariableDefinitions.docx